

The ℓ -degree Turán densities of hypergraphs: infinite families and their finite subfamilies

Hao Lin *

Guanghui Wang †

Wenling Zhou ‡

Abstract

Let $k > \ell \geq 1$ be integers. For a family $\mathcal{F}$ of k -uniform hypergraphs (k -graphs), its ℓ -degree Turán density $\gamma_\ell^{(k)}(\mathcal{F})$ is the asymptotic maximum of the normalized minimum ℓ -degree over all $\mathcal{F}$ -free k -graphs. In this paper, we prove that for all integers $k > \ell > 2$ and every $\alpha \in [0, 1)$, there exists an infinite family $\mathcal{F}$ of k -graphs such that $\gamma_\ell^{(k)}(\mathcal{F}) = \alpha$, while every finite subfamily $\mathcal{F}' \subset \mathcal{F}$ satisfies $\gamma_\ell^{(k)}(\mathcal{F}') > \alpha$. This confirms the conjecture of Mubayi and Zhao on codegree Turán densities for every $k > 3$. As an immediate consequence, the set $\Gamma_\infty^{k,\ell}$ of ℓ -degree Turán densities of nonempty k -graph families is $[0, 1)$ whenever $k > \ell > 2$, which answers a question of Lo and Markström from 2014.

Keywords: Hypergraph, minimum degree, Turán density, forbidden family.

1 Introduction

Turán-type problems ask how dense a large graph or hypergraph can be while avoiding prescribed subgraphs. For an integer $k \geq 2$, a k -uniform hypergraph, or simply a k -graph, is a pair $H = (V(H), E(H))$ with $E(H) \subseteq \binom{V(H)}{k}$. We write $v(H) = |V(H)|$ and $e(H) = |E(H)|$. Given a nonempty family $\mathcal{F}$ of k -graphs, each containing at least one edge, a k -graph is $\mathcal{F}$ -free if it contains no member of $\mathcal{F}$ as a subgraph. The classical Turán number is

$$\text{ex}(n, \mathcal{F}) = \max\{e(H) : v(H) = n, H \text{ is } \mathcal{F}\text{-free}\},$$

and the Turán density of $\mathcal{F}$ is

$$\pi(\mathcal{F}) = \lim_{n \rightarrow \infty} \frac{\text{ex}(n, \mathcal{F})}{\binom{n}{k}}.$$

The limit exists because the normalized Turán numbers form a non-increasing sequence. If $\mathcal{F} = \{F\}$, we write $\text{ex}(n, F)$ and $\pi(F)$.

Let $\Pi^{(k)}$, $\Pi_{\text{fin}}^{(k)}$ and $\Pi_\infty^{(k)}$ denote the sets of Turán densities of, respectively, single, finite families and arbitrary nonempty forbidden families of k -graphs. Clearly,

$$\Pi^{(k)} \subseteq \Pi_{\text{fin}}^{(k)} \subseteq \Pi_\infty^{(k)} \subseteq [0, 1).$$

For graphs, the Erdős–Stone–Simonovits Theorem [3, 4] determines the Turán density of every graph in terms of its chromatic number, and consequently

$$\Pi^{(2)} = \Pi_{\text{fin}}^{(2)} = \Pi_\infty^{(2)} = \{(r-1)/r : r \in \mathbb{N}^+\}. \quad (1.1)$$

*School of Mathematical Sciences, University of Science and Technology of China, Hefei, Anhui 230026, China. Email: haolinz6@ustc.edu.cn.

†School of Mathematics, Shandong University, Jinan, Shandong 250100, China. Email: ghwang@sdu.edu.cn.

‡School of Mathematics, Shandong University, Jinan, Shandong 250100, China. Email: gracezhou@sdu.edu.cn.

However, for $k \geq 3$, we do not have an analogue to the Erdős-Stone-Simonovits Theorem, and much less is known about the Turán densities of k -graphs. Turán in his original work [18] asked for the Turán number of complete k -graphs. Later, Erdős offered a monetary prize for determining $\pi(K_r^{(k)})$ for a complete k -graph $K_r^{(k)}$ with $r > k \geq 3$. To this day, no case of this problem with $r > k \geq 3$ has been resolved, even asymptotically, despite substantial effort. For further results along this line of research (when $|\mathcal{F}| = 1$), we refer to see the surveys of Keevash [8] and Sidorenko [16].

A different line of research investigates properties of the set of hypergraph Turán densities. In 1984, Frankl and Rödl [5] disproved a \$1000 conjecture of Erdős by showing that, for $k \geq 3$, the set $\Pi_\infty^{(k)}$ is not well-ordered. Later, Pikhurko [12] proved a collection of fundamental results regarding the sets $\Pi_\infty^{(k)}$ and $\Pi_{\text{fin}}^{(k)}$. In particular, he showed that $\Pi_\infty^{(k)} \neq \Pi_{\text{fin}}^{(k)}$, the set $\Pi_\infty^{(k)}$ is uncountable.

1.1 ℓ -degree Turán densities

A natural local refinement of the classical Turán problem replaces the total number of edges by a minimum ℓ -degree condition. Let $1 \leq \ell < k$, let H be a k -graph, and let $S \in \binom{V(H)}{\ell}$. The *link* of S is

$$N_H(S) = \left\{ T \in \binom{V(H) \setminus S}{k-\ell} : S \cup T \in E(H) \right\},$$

its *degree* is $d_H(S) = |N_H(S)|$, and the minimum ℓ -degree of H is $\delta_\ell(H) := \min\{d_H(S) : S \in \binom{V(H)}{\ell}\}$. For a family $\mathcal{F}$ of k -graphs, define

$$\text{ex}_\ell^{(k)}(n, \mathcal{F}) = \max\{\delta_\ell(H) : v(H) = n, H \text{ is } \mathcal{F}\text{-free}\}.$$

The corresponding ℓ -degree Turán density is

$$\gamma_\ell^{(k)}(\mathcal{F}) = \lim_{n \rightarrow \infty} \frac{\text{ex}_\ell^{(k)}(n, \mathcal{F})}{\binom{n-\ell}{k-\ell}}. \quad (1.2)$$

The limit exists for arbitrary forbidden families: Mubayi and Zhao [15] proved this in the codegree case $\ell = k - 1$, and Lo and Markström [11] proved it for all $1 \leq \ell < k$. We use the convention $\gamma_\ell^{(k)}(\emptyset) = 1$. For every forbidden family $\mathcal{F}$,

$$\gamma_{k-1}^{(k)}(\mathcal{F}) \leq \gamma_{k-2}^{(k)}(\mathcal{F}) \leq \cdots \leq \gamma_1^{(k)}(\mathcal{F}) = \pi(\mathcal{F}). \quad (1.3)$$

As in the classical setting, let $\Gamma^{k,\ell}$, $\Gamma_{\text{fin}}^{k,\ell}$ and $\Gamma_\infty^{k,\ell}$ denote the sets of ℓ -degree Turán densities obtained from a single forbidden k -graph, a finite forbidden family and an arbitrary nonempty forbidden family, respectively. Thus,

$$\Gamma^{k,\ell} \subseteq \Gamma_{\text{fin}}^{k,\ell} \subseteq \Gamma_\infty^{k,\ell} \subseteq [0, 1). \quad (1.4)$$

The first two sets in (1.4) are countable, whereas the third may contain uncountably many values. For $k = 2$ this again gives the set in (1.1). By contrast, Mubayi and Zhao [15] proved that $\Gamma_\infty^{k,k-1}$ is dense in $[0, 1)$ for every $k \geq 3$ and posed the following conjecture.

Conjecture 1.1 ([15, Conjecture 1.7]). For each $k \geq 3$ and $\alpha \in [0, 1)$, there exists an infinite family $\mathcal{F}$ of k -graphs such that $\gamma_{k-1}^{(k)}(\mathcal{F}) = \alpha$ and all finite families $\mathcal{F}' \subset \mathcal{F}$ satisfying $\gamma_{k-1}^{(k)}(\mathcal{F}') > \alpha$.

Subsequently, Lo and Markström [11, Theorem 1.7] extended the density result of Mubayi and Zhao by proving that $\Gamma_{k,\ell}^\infty$ is dense in $[0, 1)$ for every $k > \ell > 1$, and they further posed the open question of whether $\Gamma_{\infty}^{k,\ell} = [0, 1)$.

There has recently been substantial progress for finite forbidden families. Gao, Pikhurko, Rong and Sun [6] proved that every rational number in $[0, 1)$ is the codegree Turán density of a finite family of k -graphs. Tang and Zhou [17] extended this rational-value result to every pair $k > \ell > k/2$. They also established two-element non-principal families for all $k > \ell > 1$. These results describe a substantial part of the values in $\Gamma_{\text{fin}}^{k,\ell}$ and $\Gamma_{\infty}^{k,\ell}$, but they cannot by themselves yield a full interval: there are only countably many finite forbidden families, and density of this set does not imply that every value is attained.

We believe that Conjecture 1.1 can be strengthened to hold for all integers $k > \ell > 1$. Our main result extends this conjecture for all integers $k > \ell > 2$.

Theorem 1.2. *For each $k > \ell > 2$ and $\alpha \in [0, 1)$, there exists an infinite family $\mathcal{F}$ of k -graphs such that $\gamma_{k-1}^{(k)}(\mathcal{F}) = \alpha$ and all finite families $\mathcal{F}' \subset \mathcal{F}$ satisfying $\gamma_{k-1}^{(k)}(\mathcal{F}') > \alpha$.*

For $\ell = k - 1$, Theorem 1.2 implies Conjecture 1.1 for every $k > 3$. In particular, as an immediate consequence, Theorem 1.2 also answers the question of Lo and Markström for every $k > \ell \geq 3$.

Corollary 1.3. *For all integers $k > \ell \geq 3$, one has $\Gamma_{k,\ell}^\infty = [0, 1)$.*

Very recently, while the present manuscript was being completed, we learned of independent work of Li and Liu [9]. They proved that $\Gamma_{\infty}^{k,\ell} = [0, 1)$ for all integers $k > \ell > 1$ with $(k, \ell) \neq (3, 2)$. However, Theorem 1.2 establishes a stronger conclusion in the overlapping range. For every $\alpha \in [0, 1)$, we construct an infinite antichain $\mathcal{F}$ of k -graphs, none of which has an isolated vertex, such that $\gamma_{\ell}^{(k)}(\mathcal{F}) = \alpha < \gamma_{\ell}^{(k)}(\mathcal{F}')$ for every finite subfamily $\mathcal{F}' \subseteq \mathcal{F}$.

Organization. In Section 2, we collect the terminology for monotone families and minimal forbidden k -graphs, together with the Lagrangian results used later. The cyclic multipartite construction and its matching upper and lower bounds are developed in Section 3. In Section 4 we apply the construction first to positive densities and then to zero density.

2 Preliminaries

For a finite set X , let $\binom{X}{r}$ denote the family of its r -element subsets. For a positive integer n , let $[n] = \{1, \dots, n\}$. For integers $a \geq b \geq 0$, write $(a)_b = a(a-1) \cdots (a-b+1)$, with $(a)_0 = 1$. The complete r -graph on a vertices is denoted by $K_a^{(r)}$. All asymptotic statements concern the order of the host k -graph tending to infinity. Subscripts on $O(\cdot)$ specify the parameters on which the implicit constant may depend. All auxiliary r -graphs have nonempty vertex sets, and edgeless auxiliary r -graphs are allowed.

2.1 Monotone families and minimal forbidden k -graphs

Throughout the paper, a family of k -graphs is understood to be closed under isomorphism. A family $\mathcal{P}$ is *monotone* if every subgraph of a member of $\mathcal{P}$ also belongs to $\mathcal{P}$. For a family $\mathcal{F}$ of k -graphs, let $\text{Forb}(\mathcal{F})$ denote the monotone family of all $\mathcal{F}$ -free k -graphs.

Conversely, for a monotone family $\mathcal{P}$, define its associated forbidden family by

$$\mathcal{F}(\mathcal{P}) = \{F : F \text{ is a finite } k\text{-graph and } F \notin \mathcal{P}\},$$

1 where one representative of each isomorphism type is retained. Then

$$\text{Forb}(\mathcal{F}(\mathcal{P})) = \mathcal{P}. \quad (2.1)$$

2 Indeed, every subgraph of a member of $\mathcal{P}$ again lies in $\mathcal{P}$, whereas a k -graph outside $\mathcal{P}$ is itself a
 3 member of $\mathcal{F}(\mathcal{P})$. All monotone families considered below contain the edgeless k -graph of every
 4 order. For such a family, define its ℓ -degree density by

$$d_\ell(\mathcal{P}) = \gamma_\ell^{(k)}(\mathcal{F}(\mathcal{P})) = \lim_{n \rightarrow \infty} \frac{\max\{\delta_\ell(H) : H \in \mathcal{P}, v(H) = n\}}{\binom{n-\ell}{k-\ell}}. \quad (2.2)$$

5 The uniformity k will always be clear from the context. If $\mathcal{P} \subseteq \mathcal{Q}$, then $d_\ell(\mathcal{P}) \leq d_\ell(\mathcal{Q})$. Arbitrary
 6 unions and intersections of monotone families are again monotone.

7 For a monotone family $\mathcal{P}$, let

$$\mathcal{F}_{\min}(\mathcal{P}) = \{F \notin \mathcal{P} : \text{every proper subgraph of } F \text{ belongs to } \mathcal{P}\},$$

8 again retaining one representative of each isomorphism type. The members of $\mathcal{F}_{\min}(\mathcal{P})$ are the
 9 *minimal forbidden k -graphs* for $\mathcal{P}$. The following elementary observation will be used at the end of
 10 the proof.

11 **Lemma 2.1.** *Let $\mathcal{P}$ be a monotone family that is closed under adding isolated vertices, and put*
 12 *$\alpha = d_\ell(\mathcal{P})$. Suppose that $\gamma_\ell^{(k)}(\mathcal{F}') > \alpha$ for every finite $\mathcal{F}' \subseteq \mathcal{F}(\mathcal{P})$. Then $\mathcal{F}_{\min}(\mathcal{P})$ is an infinite*
 13 *antichain of k -graphs with no isolated vertices, $\text{Forb}(\mathcal{F}_{\min}(\mathcal{P})) = \mathcal{P}$, and*

$$\gamma_\ell^{(k)}(\mathcal{F}_{\min}(\mathcal{P})) = \alpha < \gamma_\ell^{(k)}(\mathcal{F}') \quad \text{for every finite } \mathcal{F}' \subseteq \mathcal{F}_{\min}(\mathcal{P}).$$

14 *Proof.* Every finite k -graph outside $\mathcal{P}$ contains a subgraph that is minimal outside $\mathcal{P}$, and hence
 15 $\text{Forb}(\mathcal{F}_{\min}(\mathcal{P})) = \mathcal{P}$. Minimality also implies that $\mathcal{F}_{\min}(\mathcal{P})$ is an antichain. If $F \in \mathcal{F}_{\min}(\mathcal{P})$ had an
 16 isolated vertex v , then $F - v \in \mathcal{P}$, and closure under adding isolated vertices would give $F \in \mathcal{P}$, a
 17 contradiction. The density statement and the strict inequality for finite subfamilies follow from the
 18 hypothesis and the inclusion $\mathcal{F}_{\min}(\mathcal{P}) \subseteq \mathcal{F}(\mathcal{P})$. Finally, $\mathcal{F}_{\min}(\mathcal{P})$ cannot be finite, since otherwise it
 19 could itself be used as $\mathcal{F}'$ in the strict inequality. $\square$

20 We also justify the exclusion of 1 from (1.4). Choose $F \in \mathcal{F}$ and put $f = v(F)$. An $\mathcal{F}$ -free k -graph
 21 H is $K_f^{(k)}$ -free, so every f -set of vertices contains a non-edge. Double counting pairs consisting of a
 22 non-edge and an f -set containing it gives

$$\frac{e(H)}{\binom{n}{k}} \leq 1 - \binom{f}{k}^{-1} \quad (n \geq f).$$

23 On the other hand,

$$\frac{\delta_\ell(H)}{\binom{n-\ell}{k-\ell}} \leq \frac{\sum_{S \in \binom{V(H)}{\ell}} d_H(S)}{\binom{n}{\ell} \binom{n-\ell}{k-\ell}} = \frac{e(H)}{\binom{n}{k}}. \quad (2.3)$$

24 Consequently $\gamma_\ell^{(k)}(\mathcal{F}) < 1$ for every nonempty forbidden family under consideration.

2.2 Hypergraph Lagrangians and Turán densities

For a finite r -graph G , let

$$\Delta_G = \left\{ (x_v)_{v \in V(G)} : x_v \geq 0, \sum_{v \in V(G)} x_v = 1 \right\}.$$

Its normalized *Lagrangian* is

$$\lambda(G) = \max_{\mathbf{x} \in \Delta_G} r! \sum_{e \in E(G)} \prod_{v \in e} x_v. \quad (2.4)$$

The maximum exists by compactness. If r colors are sampled independently according to $\mathbf{x}$, then $r! \sum_{e \in E(G)} \prod_{v \in e} x_v$ is the probability that the sampled colors are distinct and form an edge of G . In particular, $0 \leq \lambda(G) \leq 1$.

Suppose a set of size a is colored by $V(G)$, and a_v elements receive color v . The number of r -sets whose colors form an edge of G is

$$\sum_{e \in E(G)} \prod_{v \in e} a_v \leq \frac{\lambda(G)}{r!} a^r. \quad (2.5)$$

The upper bound depends on G only through $\lambda(G)$. We also record that

$$\frac{(a)_r}{a^r} \leq \lambda(K_a^{(r)}) \leq 1 \quad \text{and hence} \quad \lambda(K_a^{(r)}) \rightarrow 1 \quad \text{as } a \rightarrow \infty. \quad (2.6)$$

The lower bound follows by assigning weight $1/a$ to every vertex.

Let $\Lambda^{(r)}$ be the set of Lagrangians of finite r -graphs, and let $\Pi_\infty^{(r)}$ denote the set of Turán densities of arbitrary families of r -graphs. Here we allow the empty forbidden family, so $1 \in \Pi_\infty^{(r)}$. The following two results on classical Turán densities are used in the proof for positive densities.

Theorem 2.2 (Pikhurko [12]). *For every integer $r \geq 3$, the closure of $\Lambda^{(r)}$ is $\Pi_\infty^{(r)}$.*

Theorem 2.3 (Liu–Pikhurko [10]). *For every integer $r \geq 3$, there is a constant $\delta_r > 0$ such that $[1 - \delta_r, 1] \subseteq \Pi_\infty^{(r)}$.*

We shall use these two theorems through the following immediate consequence, which gives approximation from below.

Lemma 2.4. *For every integer $r \geq 3$, there exists $b_r \in (0, 1)$ such that every $\tau \in (b_r, 1)$ is the supremum of the Lagrangians of finite r -graphs whose Lagrangian is strictly smaller than τ .*

Proof. Choose $b_r \in (0, 1)$ so that $[b_r, 1] \subseteq \Pi_\infty^{(r)}$, using Theorem 2.3. Fix $\tau \in (b_r, 1)$ and $\varepsilon > 0$. Choose

$$\max\{b_r, \tau - \varepsilon\} < \beta < \tau,$$

and then choose $\eta > 0$ so small that $(\beta - \eta, \beta + \eta) \subseteq (\tau - \varepsilon, \tau)$. Since $\beta \in \Pi_\infty^{(r)}$, Theorem 2.2 gives a finite r -graph G with $|\lambda(G) - \beta| < \eta$. Thus $\tau - \varepsilon < \lambda(G) < \tau$, as required. $\square$

Concentration. We use Hoeffding's inequality [7] in the following form. If $X_1, \dots, X_N$ are independent random variables with values in $[0, 1]$, then, for every $u > 0$,

$$\mathbb{P} \left(\sum_{j=1}^N X_j < \mathbb{E} \sum_{j=1}^N X_j - u \right) \leq \exp \left(-\frac{2u^2}{N} \right). \quad (2.7)$$

Only sums of indicator variables occur below. Independence is needed for the extensions of a fixed ℓ -set; the degree variables of different ℓ -sets need not be independent.

3 A cyclic multipartite construction

Throughout this section, fix integers $k > \ell \geq 3$ and put $s = k - \ell$. Let m, t be integers satisfying $m \geq k$ and $s \leq t \leq m - \ell$, and set $c_{m,t} := (t)_s / m^s = s! \binom{t}{s} / m^s$. The parameters k, ℓ, m, t remain fixed throughout the construction, whereas the auxiliary ℓ -graph may vary. We define a monotone family of k -graphs and determine its minimum ℓ -degree Turán density by matching upper and lower bounds. The constants in these bounds are independent of the auxiliary ℓ -graph, a fact that will later allow us to use a family of auxiliary ℓ -graphs.

3.1 The construction

We first define the k -graphs that underlie the construction and then pass to the monotone family formed by all of their subgraphs.

Definition 3.1 (Cyclic multipartite G -construction). Let G be a finite ℓ -graph, let V be a finite set, and let $\mathcal{V} = (V_0, \dots, V_{m-1})$ be an ordered partition of V into m parts, where empty parts are allowed. We regard all part indices as elements of $\mathbb{Z}_m$. For each $i \in \mathbb{Z}_m$, put $I_i = \{i + 1, \dots, i + t\}$ and define

$$\mathcal{T}_i = \mathcal{T}_i(\mathcal{V}) := \left\{ T \in \binom{V}{s} : T \subseteq \bigcup_{j \in I_i} V_j \text{ and } |T \cap V_j| \leq 1 \text{ for every } j \in I_i \right\}. \quad (3.1)$$

Thus, every member of $\mathcal{T}_i$ is obtained by choosing one vertex from each of s distinct parts among $V_{i+1}, \dots, V_{i+t}$.

For every $i \in \mathbb{Z}_m$ and $T \in \mathcal{T}_i$, choose a map $\phi_{i,T} : V_i \rightarrow V(G)$, and write $\Phi = (\phi_{i,T})_{i \in \mathbb{Z}_m, T \in \mathcal{T}_i}$. The maps with different indices are chosen separately; in particular, no compatibility is imposed when two indexing sets T intersect.

The k -graph $H(G; \mathcal{V}, \Phi)$ has vertex set V . A set $X \in \binom{V}{k}$ is an edge of $H(G; \mathcal{V}, \Phi)$ if and only if one of the following holds.

- (i) $|X \cap V_i| \leq \ell - 1$ for every $i \in \mathbb{Z}_m$;
- (ii) for some $i \in \mathbb{Z}_m$, there are $S \in \binom{V_i}{\ell}$ and $T \in \mathcal{T}_i$ such that $X = S \dot{\cup} T$ and $\{\phi_{i,T}(u) : u \in S\} \in E(G)$.

The two edge types in Definition 3.1 are disjoint. An edge of type (ii) contains exactly ℓ vertices in one part and at most one vertex in every other part, so the index i in its definition is unique. Moreover, the colors $\phi_{i,T}(u)$, $u \in S$, are pairwise distinct whenever the set in type (ii) is an edge of G .

Let $\mathcal{P}(G; m, t)$ be the family of all k -graphs H for which there are an ordered m -partition $\mathcal{V}$ of $V(H)$ and maps Φ as in Definition 3.1 satisfying

$$E(H) \subseteq E(H(G; \mathcal{V}, \Phi)). \quad (3.2)$$

Equivalently, $\mathcal{P}(G; m, t)$ consists of all subgraphs of the k -graphs arising from the construction above. It is a monotone family: deleting vertices amounts to restricting the parts and the relevant maps, while deleting edges leaves the same witnessing data. It is also closed under adding isolated vertices. Indeed, a new vertex may be placed in any part, after which the existing maps are extended and the maps corresponding to newly created indexing sets are chosen arbitrarily.

Every part V_i is independent in $H(G; \mathcal{V}, \Phi)$. It follows that, regardless of G , every member of $\mathcal{P}(G; m, t)$ is $K_{m(k-1)+1}^{(k)}$ -free: among any $m(k-1)+1$ vertices, one part contains at least k vertices, and these vertices do not span an edge.

If $t = s$ and all k -sets of type (ii) are included, the balanced construction reduces to the cyclic multipartite construction of Lo and Markström [11]. Here the auxiliary ℓ -graph G controls the contribution of the edges of type (ii) through its Lagrangian. Assigning a separate map $\phi_{i,T}$ to each T is important both for the averaging argument in the upper bound and for the independent random choices used in the lower bound.

3.2 The uniform upper bound

We first record a cyclic estimate for the number of members of $\mathcal{T}_i$. For nonnegative reals $z_1, \dots, z_t$, let

$$e_s(z_1, \dots, z_t) = \sum_{A \in \binom{[t]}{s}} \prod_{j \in A} z_j.$$

The number of sets in $\mathcal{T}_i$ is $e_s(|V_{i+1}|, \dots, |V_{i+t}|)$. The following lemma selects a nonempty part for which this quantity is small.

Lemma 3.2. *Let $1 \leq s \leq t < m$, and let $y_0, \dots, y_{m-1} \geq 0$ with $N = \sum_i y_i > 0$. There is an index i with $y_i > 0$ such that*

$$e_s(y_{i+1}, \dots, y_{i+t}) \leq \binom{t}{s} \left(\frac{N}{m} \right)^s. \quad (3.3)$$

Proof. Set $W_j = \sum_{h=1}^t y_{j+h}$. Since $\sum_j W_j = tN$, some j satisfies $W_j \leq tN/m$. If $y_j > 0$, choose $i = j$. Otherwise let i be the last positive-weight index preceding j in the cyclic order, and let d be the forward distance from i to j . Then $y_{i+1} = \dots = y_{i+d} = 0$. If $d \geq t$, we have $W_i = 0$. If $d < t$, every possibly nonzero term in W_i also occurs in W_j . In either case,

$$y_i > 0 \quad \text{and} \quad W_i \leq tN/m. \quad (3.4)$$

For nonnegative $z_1, \dots, z_t$, Maclaurin's inequality gives

$$e_s(z_1, \dots, z_t) \leq \binom{t}{s} \left(\frac{z_1 + \dots + z_t}{t} \right)^s. \quad (3.5)$$

Applying this inequality together with (3.4) yields (3.3). $\square$

We apply the lemma after ignoring parts of order less than $\sqrt{n}$.

Proposition 3.3. *There are constants $C = C(k, \ell, m, t)$ and $n_0 = n_0(k, \ell, m, t)$ such that, for every finite ℓ -graph G , every $n \geq n_0$ and every $H \in \mathcal{P}(G; m, t)$ with $v(H) = n$,*

$$\frac{\delta_\ell(H)}{\binom{n-\ell}{s}} \leq c_{m,t} \lambda(G) + Cn^{-1/2}. \quad (3.6)$$

In particular, C and n_0 do not depend on G .

Proof. Since H is contained in some $H(G; \mathcal{V}, \Phi)$ and adding edges cannot decrease the minimum ℓ -degree, it is enough to consider $H = H(G; \mathcal{V}, \Phi)$. Put $n_i = |V_i|$, and define

$$y_i = \begin{cases} n_i, & n_i \geq \sqrt{n}, \\ 0, & n_i < \sqrt{n}. \end{cases}$$

1 For sufficiently large n , at least one y_i is positive. The parts with $y_i = 0$ contain at most $m\sqrt{n}$
2 vertices in total. Apply [Lemma 3.2](#) to $(y_i)_{i=0}^{m-1}$. It gives an index i with $n_i \geq \sqrt{n}$ and

$$e_s(y_{i+1}, \dots, y_{i+t}) \leq \binom{t}{s} \left(\frac{n}{m}\right)^s.$$

3 Every set in $\mathcal{T}_i$ either uses only parts of size at least $\sqrt{n}$ or contains a vertex from a smaller part.
4 There are at most $m\sqrt{n}\binom{n}{s-1}$ sets of the latter kind. Hence

$$|\mathcal{T}_i| \leq \binom{t}{s} \left(\frac{n}{m}\right)^s + O_{k,m}(n^{s-1/2}). \quad (3.7)$$

5 Write $a = n_i$. For $T \in \mathcal{T}_i$ and $c \in V(G)$, let $a_c(T) = |\phi_{i,T}^{-1}(c)|$. Since $\sum_c a_c(T) = a$, [\(2.5\)](#) gives

$$|\{S \in \binom{V_i}{\ell} : S \cup T \in E(H)\}| = \sum_{e \in E(G)} \prod_{c \in e} a_c(T) \leq \frac{\lambda(G)}{\ell!} a^\ell. \quad (3.8)$$

6 An ℓ -set contained in V_i has no extension to an edge of type (i). In an edge of type (ii) containing it,
7 V_i must be the part containing ℓ vertices, so its complementary s -set lies in $\mathcal{T}_i$. We may therefore
8 sum [\(3.8\)](#) over all $T \in \mathcal{T}_i$ and average over $S \in \binom{V_i}{\ell}$ to obtain

$$\delta_\ell(H) \leq \frac{1}{\binom{a}{\ell}} \sum_{S \in \binom{V_i}{\ell}} d_H(S) \leq \lambda(G) \frac{a^\ell}{(a)_\ell} |\mathcal{T}_i|. \quad (3.9)$$

9 As $a \geq \sqrt{n}$,

$$\frac{a^\ell}{(a)_\ell} = 1 + O_k(n^{-1/2}).$$

10 Together with [\(3.7\)](#) and $\lambda(G) \leq 1$, this yields

$$\delta_\ell(H) \leq \lambda(G) \binom{t}{s} \left(\frac{n}{m}\right)^s + O_{k,m}(n^{s-1/2}).$$

11 Finally, $\binom{n-\ell}{s} = n^s/s! + O_k(n^{s-1})$, so division by this quantity proves [\(3.6\)](#). All implicit constants
12 depend only on the fixed parameters k, ℓ, m, t . $\square$

13 3.3 The matching lower bound

14 For the lower bound, we choose the maps randomly. For a fixed ℓ -set contained in one part, distinct
15 complementary s -sets use independent maps, which allows us to control all ℓ -degrees simultaneously.

16 **Proposition 3.4.** There are constants $C = C(k, \ell, m, t)$ and $n_0 = n_0(k, \ell, m, t)$ such that, for every
17 finite ℓ -graph G and every $n \geq n_0$, there is an n -vertex k -graph $H \in \mathcal{P}(G; m, t)$ satisfying

$$\frac{\delta_\ell(H)}{\binom{n-\ell}{s}} \geq c_{m,t} \lambda(G) - C \left(n^{-1} + n^{-s/2} \sqrt{\log n} \right). \quad (3.10)$$

18 *Proof.* Choose a partition with $|V_i| = n/m + O(1)$ for each i , and let $\mathbf{x} \in \Delta_G$ attain $\lambda(G)$.
19 Independently for all triples (i, T, u) with $T \in \mathcal{T}_i$ and $u \in V_i$, choose $\phi_{i,T}(u)$ according to $\mathbf{x}$.
20 Let $H = H(G; \mathcal{V}, \Phi)$.

1 First fix $S \in \binom{V_i}{\ell}$. For any $T \in \mathcal{T}_i$, the probability that the colors on S form an edge of G is
2 $\lambda(G)$. These events are independent for distinct T , even if those sets share vertices, because the
3 random choices are indexed by the set T . Thus

$$d_H(S) \sim \text{Bin}(B_i, \lambda(G)), \quad B_i = |\mathcal{T}_i| = \binom{t}{s} \left(\frac{n}{m}\right)^s + O_{k,m}(n^{s-1}). \quad (3.11)$$

4 Set $u = \sqrt{\ell + 3} n^{s/2} \sqrt{\log n}$. Since $B_i \leq n^s$, (2.7) gives

$$\mathbb{P}(d_H(S) < \lambda(G)B_i - u) \leq n^{-2\ell-6}.$$

5 There are at most n^ℓ such sets S . A union bound therefore shows that, with probability tending to
6 one, all of them satisfy

$$d_H(S) \geq \lambda(G) \binom{t}{s} \left(\frac{n}{m}\right)^s - O_{k,m}(n^{s-1} + n^{s/2} \sqrt{\log n}). \quad (3.12)$$

7 The probability estimate is independent of G .

8 Next let $S \in \binom{V}{\ell}$ meet exactly $h \geq 2$ parts. Then $h \leq \ell$ and $|S \cap V_i| \leq \ell - 1$ for every i . Choose
9 s distinct parts not met by S , and take one vertex from each to obtain T . Every such $S \cup T$ is an
10 edge of type (i). There are at least

$$\begin{aligned} \binom{m-h}{s} \left(\frac{n}{m}\right)^s - O_{k,m}(n^{s-1}) &\geq \binom{m-\ell}{s} \left(\frac{n}{m}\right)^s - O_{k,m}(n^{s-1}) \\ &\geq \binom{t}{s} \left(\frac{n}{m}\right)^s - O_{k,m}(n^{s-1}) \end{aligned} \quad (3.13)$$

11 such sets T . The last inequality uses $t \leq m - \ell$. Since $\lambda(G) \leq 1$, this deterministic bound is at least
12 the required lower estimate.

13 Choose an outcome for which (3.12) holds simultaneously for every ℓ -set in a part. Combining it
14 with (3.13) and dividing by $\binom{n-\ell}{s}$ proves (3.10). $\square$

15 The two propositions determine the ℓ -degree density of the monotone family $\mathcal{P}(G; m, t)$. Since
16 the estimate in Proposition 3.3 is uniform in G , we may also allow the auxiliary ℓ -graph to range
17 over an arbitrary family.

18 3.4 Auxiliary ℓ -graphs

19 We now allow the auxiliary ℓ -graph to range over a nonempty family. Throughout this subsection,
20 $s = k - \ell$ and the integers m, t satisfy $m \geq k$ and $s \leq t \leq m - \ell$.

21 **Proposition 3.5.** Let $k > \ell \geq 3$, and let $\mathcal{G}$ be a nonempty family of finite ℓ -graphs. Set $\tau :=$
22 $\sup_{G \in \mathcal{G}} \lambda(G)$ and $\mathcal{P}(\mathcal{G}; m, t) := \bigcup_{G \in \mathcal{G}} \mathcal{P}(G; m, t)$. Then $d_\ell(\mathcal{P}(\mathcal{G}; m, t)) = c_{m,t}\tau$. In particular, the
23 associated forbidden family is nonempty and has ℓ -degree Turán density $c_{m,t}\tau$.

24 *Proof.* The union $\mathcal{P} = \mathcal{P}(\mathcal{G}; m, t)$ is a monotone family and contains the edgeless k -graph of every
25 order. It excludes $K_{m(k-1)+1}^{(k)}$, so $\mathcal{F}(\mathcal{P})$ is nonempty. For any n -vertex $H \in \mathcal{P}$, there is some $G \in \mathcal{G}$
26 with $H \in \mathcal{P}(G; m, t)$. By Proposition 3.3,

$$\frac{\delta_\ell(H)}{\binom{n-\ell}{s}} \leq c_{m,t}\lambda(G) + Cn^{-1/2} \leq c_{m,t}\tau + Cn^{-1/2}.$$

The same constant C and the same lower bound on n apply to every $G \in \mathcal{G}$. This proves $d_\ell(\mathcal{P}) \leq c_{m,t}\tau$ even though the choice of G may depend on H and on n .

For the reverse inequality, fix $\varepsilon > 0$ and choose $G \in \mathcal{G}$ with $\lambda(G) > \tau - \varepsilon$. By [Proposition 3.4](#), for every sufficiently large n there is an $H \in \mathcal{P}(G; m, t) \subseteq \mathcal{P}$ with

$$\frac{\delta_\ell(H)}{\binom{n-\ell}{s}} \geq c_{m,t}(\tau - \varepsilon) - o(1).$$

Letting $n \rightarrow \infty$ and then $\varepsilon \rightarrow 0$ proves the claimed equality. $\square$

The conclusion of [Proposition 3.5](#) does not follow from the individual densities alone: in general, the density of an increasing union of monotone families need not equal the supremum of their densities. Here [Proposition 3.3](#) gives, for every n , an upper bound that is uniform over all $G \in \mathcal{G}$, and this is the additional information needed for the union.

We next record that the coefficients $c_{m,t}$ are dense in $(0, 1)$.

Lemma 3.6. *For fixed $k > \ell \geq 3$ and $s = k - \ell$, the set*

$$\left\{ \frac{\binom{t}{s}}{m^s} : m \geq k, s \leq t \leq m - \ell \right\}$$

is dense in $(0, 1)$.

Proof. Fix $\beta \in (0, 1)$ and put $q = \beta^{1/s}$. For each sufficiently large m , let $t = \lfloor qm \rfloor$. Since $0 < q < 1$, we have $s \leq t \leq m - \ell$ for all sufficiently large m . Moreover,

$$\frac{\binom{t}{s}}{m^s} = \prod_{j=0}^{s-1} \frac{\lfloor qm \rfloor - j}{m} \longrightarrow q^s = \beta.$$

$\square$

4 Proof of the main theorem

Throughout this section, put $s = k - \ell$. We construct a monotone family $\mathcal{Q}$ with $d_\ell(\mathcal{Q}) = \alpha$ such that every finite subfamily of its associated forbidden family $\mathcal{F}(\mathcal{Q})$ has strictly larger ℓ -degree Turán density. For positive α , the integers m, t remain fixed while the upper bound on the auxiliary Lagrangian decreases to the required value. For $\alpha = 0$, we intersect a sequence of monotone families arising from uncolored cyclic multipartite constructions. We then apply [Lemma 2.1](#) to obtain the antichain in the statement of the theorem.

4.1 Positive densities

Proposition 4.1. Let $k > \ell \geq 3$ and $0 < \alpha < 1$. There exists a monotone family $\mathcal{Q}$, closed under adding isolated vertices, such that $d_\ell(\mathcal{Q}) = \alpha$ and $\gamma_\ell^{(k)}(\mathcal{F}') > \alpha$ for every finite $\mathcal{F}' \subseteq \mathcal{F}(\mathcal{Q})$.

Proof. Let $b_\ell \in (0, 1)$ be given by [Lemma 2.4](#). Since $0 < \alpha < 1$, the interval $(\alpha, \min\{1, \alpha/b_\ell\})$ is nonempty. By [Lemma 3.6](#), choose integers m, t satisfying $m \geq k$ and $s \leq t \leq m - \ell$ such that, with $c = c_{m,t}$,

$$\alpha < c < \min\{1, \alpha/b_\ell\}. \tag{4.1}$$

1 Put $\tau = \alpha/c$. Then $b_\ell < \tau < 1$. For each $u \in (b_\ell, 1)$, define

$$\mathcal{Q}(u) = \bigcup_{\substack{G \text{ a finite } \ell\text{-graph} \\ \lambda(G) < u}} \mathcal{P}(G; m, t). \quad (4.2)$$

2 Each $\mathcal{Q}(u)$ is a monotone family and is closed under adding isolated vertices. By [Lemma 2.4](#)
3 and [Proposition 3.5](#),

$$d_\ell(\mathcal{Q}(u)) = cu. \quad (4.3)$$

4 Also, $u < v$ implies $\mathcal{Q}(u) \subseteq \mathcal{Q}(v)$.

5 Take

$$\tau_i = \tau + \frac{1 - \tau}{i + 1} \quad (i \geq 1), \quad \mathcal{Q} = \bigcap_{i \geq 1} \mathcal{Q}(\tau_i).$$

6 Then $\tau_i \downarrow \tau$, and

$$\mathcal{Q}(\tau) \subseteq \mathcal{Q} \subseteq \mathcal{Q}(\tau_i) \quad \text{for every } i.$$

7 Using (4.3) and monotonicity gives

$$\alpha = c\tau \leq d_\ell(\mathcal{Q}) \leq \inf_{i \geq 1} c\tau_i = c\tau = \alpha.$$

8 Thus $d_\ell(\mathcal{Q}) = \alpha$.

9 Let $\mathcal{F}_i = \mathcal{F}(\mathcal{Q}(\tau_i))$. Since the families $\mathcal{Q}(\tau_i)$ form a decreasing sequence, the forbidden families
10 $\mathcal{F}_i$ form an increasing sequence. Furthermore,

$$\mathcal{F}(\mathcal{Q}) = \bigcup_{i \geq 1} \mathcal{F}_i. \quad (4.4)$$

11 Indeed, a finite k -graph is outside $\mathcal{Q}$ if and only if it is outside at least one $\mathcal{Q}(\tau_i)$.

12 Fix a nonempty finite family $\mathcal{F}' \subseteq \mathcal{F}(\mathcal{Q})$. By (4.4) and the nesting, there is an i such that
13 $\mathcal{F}' \subseteq \mathcal{F}_i$. Forbidding fewer k -graphs cannot lower the density, so

$$\gamma_\ell^{(k)}(\mathcal{F}') \geq \gamma_\ell^{(k)}(\mathcal{F}_i) = d_\ell(\mathcal{Q}(\tau_i)) = c\tau_i > c\tau = \alpha.$$

14 The assertion for the empty subfamily follows from the convention $\gamma_\ell^{(k)}(\emptyset) = 1$. Finally, since every
15 family in (4.2) is closed under adding isolated vertices, their intersection $\mathcal{Q}$ has the same closure
16 condition. $\square$

17 4.2 Zero density

18 For zero density, no auxiliary ℓ -graph is needed. We use the uncolored cyclic multipartite construction
19 and show that finitely many such constructions can be realized simultaneously.

20 Fix $k > \ell \geq 3$, and again put $s = k - \ell$. For $m \geq k$ and a partition $\mathcal{V} = (V_0, \dots, V_{m-1})$, let
21 $J_m(\mathcal{V})$ be the k -graph whose edges are

- 22 (i) all k -sets meeting each part in at most $\ell - 1$ vertices;
- 23 (ii) all $S \cup T$ with $S \in \binom{V_i}{\ell}$, $T \in \binom{V_s}{s}$, and $|T \cap V_{i+h}| = 1$ for each $h \in [s]$.

Here all part indices are modulo m . Let $\mathcal{R}_m$ be the monotone family consisting of all subgraphs of the k -graphs $J_m(\mathcal{V})$, where the part sizes are arbitrary. The balanced members of $\mathcal{R}_m$ are the cyclic constructions discussed in [Section 3](#).

We have the following identity of monotone families:

$$\mathcal{R}_m = \bigcup_{a \geq \ell} \mathcal{P}(K_a^{(\ell)}; m, s). \quad (4.5)$$

One inclusion is immediate by ignoring the color restrictions. For the other, given a finite partition $\mathcal{V}$, choose $a \geq \max\{\ell, |V_0|, \dots, |V_{m-1}|\}$ and color every part injectively by $V(K_a^{(\ell)})$. Use that coloring for each map indexed by T . All edges of type (ii) are then present. By [\(2.6\)](#) and [Proposition 3.5](#),

$$d_\ell(\mathcal{R}_m) = \frac{s!}{m^s}. \quad (4.6)$$

In particular, these positive densities tend to zero as $m \rightarrow \infty$. The next lemma shows that the intersection of any finitely many of these monotone families has positive ℓ -degree density.

Lemma 4.2. *Let $q \geq 1$ and $m_1, \dots, m_q \geq k$. Then*

$$d_\ell \left(\bigcap_{j=1}^q \mathcal{R}_{m_j} \right) \geq \prod_{j=1}^q \frac{s!}{m_j^s} > 0. \quad (4.7)$$

Proof. Put $M = \prod_{j=1}^q m_j$ and partition an n -element set V into the cells of a common refinement,

$$W_{\mathbf{a}}, \quad \mathbf{a} \in \prod_{j=1}^q \{0, \dots, m_j - 1\}, \quad |W_{\mathbf{a}}| = \frac{n}{M} + O(1).$$

For every $j \in [q]$, the j th coordinate of the cells defines an m_j -part partition $\mathcal{V}^{(j)}$ by

$$V_i^{(j)} = \bigcup_{\mathbf{a}: a_j = i} W_{\mathbf{a}} \quad (0 \leq i < m_j).$$

On the common vertex set V , form $J_j = J_{m_j}(\mathcal{V}^{(j)})$ and define H by

$$E(H) = \bigcap_{j=1}^q E(J_j).$$

Then $H \in \bigcap_{j=1}^q \mathcal{R}_{m_j}$.

Fix $S \in \binom{V}{\ell}$. For each j , let h_j be the number of parts of $\mathcal{V}^{(j)}$ met by S . We count ordered lists $(v_1, \dots, v_s)$ of extension vertices, imposing the following requirements separately in each coordinate j .

If $h_j = 1$, say $S \subseteq V_i^{(j)}$, require the s vertices to occupy the parts $V_{i+1}^{(j)}, \dots, V_{i+s}^{(j)}$, one in each. There are $s!$ assignments of these part indices to the ordered vertices. If $h_j \geq 2$, require their part indices to be distinct and to avoid all parts met by S . There are $(m_j - h_j)_s$ assignments. Write

$$A_j(S) = \begin{cases} s!, & h_j = 1, \\ (m_j - h_j)_s, & h_j \geq 2. \end{cases}$$

Since $h_j \leq \ell$ and $m_j \geq k = \ell + s$, in both cases $A_j(S) \geq s!$.

There are $\prod_j A_j(S)$ choices for the coordinate assignments of the ordered vertices. Each choice specifies s distinct cells W_a . These cells are disjoint from S : in every coordinate their indices avoid the parts occupied by S . Thus each assignment yields

$$\left(\frac{n}{M}\right)^s + O_{k,m_1,\dots,m_q}(n^{s-1})$$

ordered lists of distinct vertices outside S . Every resulting set $S \cup \{v_1, \dots, v_s\}$ is an edge of J_j for each j . It is of type (ii) when $h_j = 1$ and of type (i) when $h_j \geq 2$.

The requirements are invariant under permuting the extension vertices. Consequently each unordered extension is counted exactly $s!$ times. It follows that

$$\begin{aligned} d_H(S) &\geq \frac{n^s}{s!} \prod_{j=1}^q \frac{A_j(S)}{m_j^s} - O_{k,m_1,\dots,m_q}(n^{s-1}) \\ &\geq \frac{n^s}{s!} \prod_{j=1}^q \frac{s!}{m_j^s} - O_{k,m_1,\dots,m_q}(n^{s-1}). \end{aligned} \quad (4.8)$$

The estimate is uniform in S . Dividing by $\binom{n-\ell}{s}$ and letting $n \rightarrow \infty$ proves (4.7). $\square$

The conclusion of Lemma 4.2 uses the particular form of these cyclic multipartite families. Its proof constructs a single k -graph belonging to all of them; the lower bound does not follow merely from their individual densities.

We now intersect the families $\mathcal{R}_m$.

Proposition 4.3. For every $k > \ell \geq 3$, there exists a monotone family $\mathcal{Q}$, closed under adding isolated vertices, such that $d_\ell(\mathcal{Q}) = 0$ and $\gamma_\ell^{(k)}(\mathcal{F}') > 0$ for every finite $\mathcal{F}' \subseteq \mathcal{F}(\mathcal{Q})$.

Proof. Let

$$\mathcal{Q} = \bigcap_{m \geq k} \mathcal{R}_m.$$

This is a monotone family containing the edgeless k -graph of every order. By (4.6) and monotonicity,

$$0 \leq d_\ell(\mathcal{Q}) \leq \inf_{m \geq k} d_\ell(\mathcal{R}_m) = \inf_{m \geq k} \frac{s!}{m^s} = 0.$$

Fix a nonempty finite $\mathcal{F}' \subseteq \mathcal{F}(\mathcal{Q})$. For each $F \in \mathcal{F}'$, choose $m(F) \geq k$ such that $F \notin \mathcal{R}_{m(F)}$. Set $M = \max_{F \in \mathcal{F}'} m(F)$. Every member of $\bigcap_{m=k}^M \mathcal{R}_m$ is $\mathcal{F}'$ -free: otherwise it would contain some $F \in \mathcal{F}'$ and, by monotonicity of $\mathcal{R}_{m(F)}$, would force $F \in \mathcal{R}_{m(F)}$. Applying Lemma 4.2 to the finite sequence $k, k+1, \dots, M$, we conclude that

$$\gamma_\ell^{(k)}(\mathcal{F}') \geq d_\ell \left(\bigcap_{m=k}^M \mathcal{R}_m \right) \geq \prod_{m=k}^M \frac{s!}{m^s} > 0.$$

Again, the empty subfamily has density 1. Closure under adding isolated vertices holds for each $\mathcal{R}_m$, by its definition or by (4.5), and hence also for $\mathcal{Q}$. $\square$

Proof of Theorem 1.2. For $\alpha > 0$, choose the monotone family $\mathcal{Q}$ supplied by Proposition 4.1; for $\alpha = 0$, use Proposition 4.3. The hypotheses of Lemma 2.1 are satisfied in either case. Applying that lemma to $\mathcal{Q}$ produces an infinite antichain $\mathcal{F} = \mathcal{F}_{\min}(\mathcal{Q})$ of k -graphs without isolated vertices such that $\gamma_\ell^{(k)}(\mathcal{F}) = \alpha$ and every finite subfamily has strictly larger minimum ℓ -degree Turán density. $\square$

Declaration on the use of AI

The authors used AI for language editing and exploratory discussion of proof presentation. All mathematical statements, proofs, and references were independently written and finalized by the authors, who take full responsibility for the content of the paper.

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